

FRC Research — Week 3 Working Notes

Framework v0.1, Smoke Test, and Baseline Reproduction

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September 22, 2026

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1. Purpose of These Notes

These are living research notes for the third week of the FRC project.

Week 2 completed the takeover and characterization of the inherited WarpX-based formation workflow. The inherited 2000-step case is now sufficiently understood to serve as a quantitative regression reference.

The focus of Week 3 therefore changes from

understanding inherited work

to

building an independently controlled and reproducible FRC workflow

without changing the validated baseline physics.

The principal objective of this week is to construct the first minimal version of the FRC simulation framework, verify the complete execution pipeline with a small smoke test, and then reproduce the inherited 2000-step baseline using the new workflow.

2. Handoff from Week 2

The takeover audit established the following working interpretation of the inherited formation case:

- WarpX is used in 3D hybrid-PIC mode with MPI and OpenMP on CPU.
- The inherited reference simulation evolves for 2000 steps with

$$\Delta t = 1 \text{ ns},$$

corresponding to

$$t_{\text{end}} = 2 \text{ } \mu\text{s}.$$

- The imposed magnetic-field rise time is

$$t_{\text{peak}} = 4 \text{ } \mu\text{s},$$

so the inherited case terminates before completion of the full field ramp.

- Two reference realizations, i and j, use the same macroscopic physical and numerical inputs but different stochastic PIC particle realizations.
- The two runs are not bitwise identical, but their macroscopic magnetic and density evolution is highly reproducible.
- A coherent reversed-field structure develops during the available interval.
- Strong compression occurs near

$$t \approx 1.7 \text{ } \mu\text{s},$$

followed by partial magnetic relaxation by

$$t = 2 \text{ } \mu\text{s}.$$

The inherited case is therefore classified as a

computational regression baseline

rather than the final scientific formation simulation.

The longer scientific formation case will be defined only after the new framework reproduces this reference successfully.

3. Week 3 Objective

The main Week 3 objective is:

Build Framework v0.1, execute a minimal smoke test, and reproduce the inherited 2000-step formation baseline without changing its physical model.

The intended progression is

Framework v0.1 → Smoke Test → 2000-Step Regression → Scientific Formation Case
--

The first three stages belong to verification and software engineering.

The final stage begins the next physical study.

4. Development Rules for This Phase

The following rules are used to avoid mixing software refactoring with changes to the physical model.

1. The inherited reference remains read-only.

The directories

~/WQ/FRC ~/WQ/WarpX

are treated as legacy reference environments.

No `git reset`, `git clean`, `git pull`, `rebuild`, or direct editing should be performed there during baseline reproduction.

2. Framework development occurs independently.

The working area for this phase is

~/WQ/FRC_jpsr/frc_framework_v0_1

together with the previously created analysis tools under

~/WQ/FRC_jpsr/analysis

3. Physics and refactoring are not changed simultaneously.

The first full regression run must preserve the inherited physics and numerical model.

4. Generated WarpX inputs must remain readable.

The framework should organize and validate the model, not hide WarpX behind an opaque software layer.

5. Every new run must be self-describing.

Configuration, executable provenance, Slurm parameters, generated inputs, and output paths should be recoverable after the run finishes.

5. Framework v0.1 Scope

Framework v0.1 is deliberately minimal.

It is not intended to be a complete simulation platform yet.

Its first responsibility is to provide a controlled pipeline

```

structured case configuration
  |
  v
validation
  |
  v
generated WarpX input
  |
  v
generated Slurm job
  |
  v
WarpX
  |
  v
recorded outputs
  |
  v
run manifest

```

The framework should initially support only the inherited FRC formation workflow.

Tilt, Grad-Shafranov initialization, compression studies, parameter sweeps, and more advanced modules can be added after the formation regression baseline is reproduced.

6. Initial Framework Structure

The target structure for the first implementation is

```

frc_framework_v0_1/
|-- configs/
|   '-- formation/
|       |-- baseline.toml
|       '-- smoke.toml
|
|-- src/
|   '-- frc/
|       |-- __init__.py
|       |-- config.py
|       |-- validation.py
|       |-- input_generator.py
|       '-- manifest.py
|
|-- templates/
|   |-- warpx_formation.in
|   '-- scnet_cpu.sbatch
|
|-- scripts/
|   |-- generate_case.py
|   '-- submit_case.py
|
|-- runs/
|-- tests/

```

```
|-- docs/
|
'-- reference/
    '-- legacy_baseline/
        |-- input_test
        |-- run.sbatch
        |-- warpx_used_inputs_i
        '-- warpx_used_inputs_j
```

The `reference/legacy_baseline` files are snapshots used for comparison and should not be edited during framework development.

7. Configuration Design

The case configuration will become the single source of truth for each simulation.

The first version will use TOML because it is readable, explicit, and suitable for separating related parameter groups.

The configuration should distinguish at least the following categories.

7.1. Case Metadata

Examples:

- case name,
- description,
- parent/reference case,
- intended purpose such as smoke or regression.

7.2. Physics

Examples:

- initial density,
- electron temperature,
- ion temperature,
- bias magnetic field,
- driven magnetic field,
- field rise time,
- ion species and mass,
- resistive and hyper-resistive model parameters.

7.3. Geometry

Examples:

- chamber radius,
- physical axial length,
- simulation-domain dimensions,
- embedded-boundary parameters,
- PML configuration.

7.4. Numerics

Examples:

- grid resolution,
- timestep,
- total number of steps,
- particles per cell,
- particle shape,
- filters,
- hybrid-solver substeps,
- diagnostic cadence.

7.5. Runtime and HPC

Examples:

- WarpX executable,
- SCNet partition,
- number of nodes,
- MPI ranks,
- OpenMP threads,
- walltime,
- checkpoint behavior.

The long-term objective is that physical parameter studies can be performed by changing a case configuration rather than manually editing multiple scripts.

8. Validation Philosophy

Framework v0.1 should catch simple but scientifically important inconsistencies before a simulation is submitted.

The most immediate example comes directly from the inherited baseline.

The total physical runtime is

$$t_{\text{end}} = N_t \Delta t.$$

For the inherited case,

$$N_t = 2000, \quad \Delta t = 1 \text{ ns},$$

so

$$t_{\text{end}} = 2 \text{ } \mu\text{s}.$$

Since

$$t_{\text{peak}} = 4 \text{ } \mu\text{s},$$

the framework should automatically detect

$$t_{\text{end}} < t_{\text{peak}}$$

and produce a warning similar to

```
WARNING:
simulation end time = 2.0 us
field-ramp peak time = 4.0 us
This case terminates before completion of the imposed field ramp.
```

This warning should not prevent execution because the 2000-step case is a deliberate regression reference.

It should, however, prevent this condition from remaining scientifically implicit.

Additional first-version checks should include:

- positive grid dimensions,
- positive timestep,
- positive particle count,
- positive number of timesteps,
- diagnostic intervals compatible with the total run length,
- valid MPI/OpenMP resource counts,
- explicit output directory,
- detection of accidental case duplication where practical.

9. Run Provenance

Each independently executed case should create a run manifest.

A minimal manifest should eventually record:

- case name,
- creation time,
- source configuration,
- generated WarpX input,
- WarpX version and commit,
- WarpX executable checksum,
- framework version or Git commit,
- SCNet partition,
- Slurm job ID,
- node count,
- MPI ranks,
- OpenMP threads,
- walltime request,
- output directory,
- and reference/parent case.

The goal is that a future result can be traced from a figure or run directory back to the exact simulation configuration that produced it.

10. Smoke Test

The first independent execution under Framework v0.1 will be a smoke test.

The smoke test is **not** intended to validate FRC physics.

Its only purpose is to verify that the complete execution path works:

config → validation → WarpX input → Slurm → WarpX → output

The smoke case should preserve the essential model configuration while reducing computational and storage cost.

The initial target is approximately

10–20

timesteps, with minimal diagnostics and without the large full checkpoints used in the inherited production run.

A successful smoke test should demonstrate:

- the generated WarpX input is valid,
- the inherited WarpX executable can run from the new workspace,

- output paths are correctly isolated,
- the Slurm environment is correct,
- the manifest is created,
- and the resulting output can be read by the existing analysis tools.

11. 2000-Step Regression Run

After the smoke test succeeds, the inherited 2000-step case will be reproduced through the new framework.

This run is the principal verification milestone for Framework v0.1.

Because the particle initialization is stochastic, successful reproduction does not require byte-for-byte equality with either inherited reference run.

Instead, the new run should reproduce the established macroscopic behavior.

The principal regression observables are

$$B_{z,\text{center}}(t),$$

$$B_{z,\text{min}}(t),$$

$$B_{z,\text{max}}(t),$$

$$\rho_{\text{center}}(t),$$

and

$$\rho_{\text{max}}(t).$$

Spatial diagnostics should also reproduce the qualitative development of the reversed-field region.

The expected reference sequence is

initial bias $\rightarrow$ field reversal $\rightarrow$ strong compression $\rightarrow$ partial relaxation

with the strongest compression occurring near

$$t \approx 1.7 \mu\text{s}.$$

Pointwise quantities such as

$$T_{i,\text{max}}$$

should not initially be used as strict pass/fail criteria because they show substantially greater sensitivity to local PIC noise.

12. After Regression: Scientific Formation Baseline

Only after the 2000-step regression succeeds should the physical problem be extended.

The first scientific formation case should evolve through at least

$$t_{\text{peak}} = 4 \mu\text{S},$$

and probably beyond this time.

This longer case will be used to determine what happens after completion of the external field ramp and to begin quantitative analysis of

- magnetic topology,
- flux retention,
- density and temperature evolution,
- particle losses,
- energy evolution,
- FRC lifetime,
- and later stability.

At that stage, reconstruction of the poloidal flux function

$$\psi(r, z)$$

should be developed in order to identify the true separatrix and derive quantities such as

$$R_s, \quad L_s, \quad E = \frac{L_s}{2R_s}.$$

13. Week 3 Milestones

The intended Week 3 milestones are:

- ☐ Create the Framework v0.1 directory structure.
- ☐ Freeze the minimal inherited reference files required for regression.
- ☐ Define `baseline.toml` as the structured representation of the inherited 2000-step case.
- ☐ Derive `smoke.toml` from the baseline while preserving the physical model.
- ☐ Implement configuration loading.
- ☐ Implement first validation rules.
- ☐ Implement readable WarpX input generation.
- ☐ Implement Slurm job generation/submission.
- ☐ Implement run-manifest creation.
- ☐ Execute the first smoke test.
- ☐ Inspect and validate smoke-test output.

- ☐ Execute the independent 2000-step regression run.
- ☐ Compare the new run against inherited references `i` and `j`.
- ☐ Document any differences before introducing optimization or new physics.

14. Current Research Log

2026-09-22 — Start of Week 3

The Week 2 takeover audit is considered complete.

The inherited 2000-step FRC formation case has been characterized both computationally and physically and can now serve as the first regression reference.

The project focus has shifted to Framework v0.1.

The immediate next task is to create and verify the framework skeleton and then encode the inherited formation case in

```
configs/formation/baseline.toml
```

before implementing the corresponding smoke-test case.

Next entry

Continue with:

- Framework v0.1 skeleton verification;
- `baseline.toml`;
- `smoke.toml`;
- configuration loading and validation.

15. Research Log — 2026-09-22

The main objective of today's session was to begin the implementation of Framework v0.1 using the inherited 2000-step formation case as the regression reference.

The framework directory structure was created under

```
~/WQ/FRC_jpsr/frc_framework_v0_1
```

with separate locations for configurations, source code, templates, scripts, runs, tests, documentation, and frozen legacy-reference files.

The following inherited files were copied into the framework reference area:

```
reference/legacy_baseline/input_test
reference/legacy_baseline/run.sbatch
reference/legacy_baseline/warpx_used_inputs_i
reference/legacy_baseline/warpx_used_inputs_j
```

These files are treated as read-only reference material.

15.1. Baseline Configuration

The inherited formation case was encoded in

```
configs/formation/baseline.toml
```

The configuration separates the case into logical sections describing

- geometry,
- PML configuration,
- plasma parameters,
- magnetic drive,
- numerical parameters,
- boundary conditions,
- hybrid-PIC model parameters,
- diagnostics,
- checkpoints,
- SCNet runtime configuration,
- reference metadata,
- and regression metadata.

The first pasted version of the TOML file contained several truncated lines. Rather than repairing the file manually section by section, it was replaced with a clean copy and independently checked with a TOML parser before continuing.

The final configuration passed the syntax check successfully.

15.2. Configuration Loader and Validation Layer

The first software components of Framework v0.1 were implemented:

```
src/frc/config.py
src/frc/validation.py
scripts/validate_case.py
```

The configuration loader reads the TOML case and derives quantities that do not need to be stored explicitly in the user configuration.

For the inherited baseline, the framework derived

$$L_x = L_y = 0.7392 \text{ m},$$

$$\Delta x = \Delta y = 0.01155 \text{ m},$$

$$\Delta z = 0.0133333 \text{ m},$$

and the total axial grid including the in-domain PML as

$$N_{z,\text{total}} = 208.$$

The physical runtime was correctly reconstructed as

$$t_{\text{end}} = N_t \Delta t = 2000 \times 10^{-9} \text{ s} = 2 \text{ } \mu\text{s}.$$

The imposed magnetic-field rise time is

$$t_{\text{peak}} = 4 \text{ } \mu\text{s}.$$

Therefore, the framework correctly detected that the inherited regression case terminates before completion of the imposed field ramp.

At the final time of the regression case, the temporal drive factor is

$$\sin\left(\frac{\pi t_{\text{end}}}{2t_{\text{peak}}}\right) = \sin\left(\frac{\pi}{4}\right) \simeq 0.7071.$$

This condition is expected for the inherited regression case and is therefore reported as a warning rather than an error.

15.3. First Successful Framework Validation

The command

```
python scripts/validate_case.py \
    configs/formation/baseline.toml
```

completed successfully with

```
VALIDATION PASSED (1 warning(s))
```

The validator also confirmed

- the physical grid dimensions,
- the 208-cell axial domain including PML,
- the 32-thread execution layout,
- the inherited WarpX executable,
- and the presence of all four frozen legacy-reference files.

The intended SCNet execution layout was reconstructed as

1 node, 4 MPI ranks, 8 OpenMP threads/rank,

for a total of

32

CPU threads.

15.4. Status at End of Session

The first stage of Framework v0.1 is now working:

```
baseline.toml → configuration loader → derived quantities → validation
```

No WarpX job has yet been submitted from the new framework.

The next development step will be to implement

```
src/frc/input_generator.py
```

so that the structured TOML configuration can generate a human-readable WarpX input.

The generated input will first be compared against the frozen inherited `input_test` before any smoke-test execution is attempted.

The immediate sequence for the next session is therefore

```
generate WarpX input → compare with legacy input → define smoke case → run smoke test
```

16. Research Log — 2026-09-23

The objective of today's session was to complete the execution layer of Framework v0.1 and perform the first end-to-end smoke test on SCNet.

At the beginning of the session, the structured formation baseline could already be loaded and validated from

```
configs/formation/baseline.toml
```

and the next task was to connect this configuration to actual WarpX input generation and SCNet execution.

16.1. WarpX Input Generation

The first version of the WarpX input generator was implemented using

```
src/frc/input_generator.py
templates/warpx_formation.in
scripts/generate_case.py
```

The purpose of this layer is to transform a structured TOML case configuration into a human-readable WarpX input rather than editing WarpX input files directly.

The inherited 2000-step formation baseline was generated successfully as

```
runs/formation_baseline_legacy_2000/
  case.toml
  input
```

The generated input retained readable numerical values and reproduced the active settings of the inherited case.

A structural comparison was performed between

```
reference/legacy_baseline/input_test
```

and the newly generated WarpX input.

Both files contained

active parameter assignments.

The comparison found

```
Missing from generated:
  NONE

Extra in generated:
  NONE

STRUCTURE MATCH: PASS
```

Therefore, no active WarpX configuration key was lost or accidentally added during the transition from the inherited input file to Framework v0.1.

16.2. Smoke-Test Configuration

A dedicated workflow smoke case was then derived from the baseline:

```
configs/formation/smoke.toml
```

The smoke test deliberately preserved the same physical model, geometry, boundary conditions, field drive, timestep, hybrid-PIC solver, and transport models while reducing the computational workload.

The principal changes relative to the inherited baseline were

$$N_t : 2000 \rightarrow 20,$$

$$N_{\text{ppc}} : 200 \rightarrow 20,$$

with a diagnostic written at step 20 and checkpoints disabled.

The physical duration of the smoke test was therefore

$$t_{\text{end}} = 20(10^{-9} \text{ s}) = 2 \times 10^{-8} \text{ s} = 0.02 \text{ } \mu\text{s}.$$

The configuration validator correctly reported that this is much shorter than

$$t_{\text{peak}} = 4 \text{ } \mu\text{s},$$

and treated the incomplete field ramp as an expected workflow-test warning rather than an error.

The temporal drive amplitude at the end of the smoke test was calculated as

$$A_{\text{time}}(t_{\text{end}}) \simeq 0.007854.$$

The smoke configuration passed validation successfully.

16.3. Conditional Checkpoint Generation

The input-generation layer was extended so that the TOML option

```
[checkpoints]
enabled = false
```

actually removes the WarpX checkpoint diagnostic from the generated input.

The final smoke input contained

```
diagnostics.diams_names = diag1
# Checkpoints disabled for this case.

diag1.intervals = 20
```

and no active `chk.*` directives.

This prevents a short workflow test from accidentally producing the multi-gigabyte checkpoints observed in the inherited production runs.

16.4. Runtime and Provenance Layer

The SCNet execution and provenance layer was implemented using

```
templates/scnet_cpu.sbatch
src/frc/manifest.py
scripts/generate_case.py
scripts/submit_case.py
```

A generated runnable case now contains

```
runs/<case_name>/
  case.toml
  input
  run.sbatch
  manifest.json
  jobs/
```

The manifest stores the case metadata, validation information, derived quantities, requested SCNet resources, WarpX information, and SHA256 hashes of the generated scientific inputs and relevant framework files.

Before submission, `submit_case.py` checks these hashes. This prevents a case from being submitted if its generated input, TOML snapshot, or Slurm script has been modified after provenance was recorded.

A dry-run of the smoke case reported

```
Integrity: PASS
DRY RUN: no job submitted.
```

confirming that the complete generated case was internally consistent.

16.5. First End-to-End Framework Execution

The first simulation launched entirely through Framework v0.1 was

```
formation_smoke_20
```

with Slurm job

```
122839324
```

on node

```
a02r1n14
```

using the inherited CPU layout

$4 \text{ MPI ranks} \times 8 \text{ OpenMP threads/rank} = 32 \text{ allocated CPU threads.}$

Slurm reported

```
State      = COMPLETED
Elapsed    = 00:01:13
ExitCode    = 0:0
```

and the framework runtime record ended with

```
EXIT_CODE=0
```

WarpX executed all 20 requested steps successfully.
The job produced

```
diag000000/  
diag000020/  
runtime.txt  
warpX_used_inputs
```

and produced no checkpoint directory.
Therefore, the complete workflow

```
TOML → validation → WarpX input → manifest → Slurm → MPI/OpenMP → WarpX → diagnostics
```

was successfully demonstrated.
The result is considered

```
Framework v0.1 smoke test: PASS
```

This test validates the execution pipeline only; it is not a physics validation of FRC formation.

16.6. Legacy Parameters Reported as Unused

WarpX reported several inherited parameters as unused, including

```
my_constants.L_r  
my_constants.R_mirror  
my_constants.mirror_factor  
my_constants.etah_vac  
my_constants.kB  
diag1.openpmd_encoding
```

This is consistent with the previous audit: several quantities remain in the legacy input even though the currently active field and hyper-resistivity expressions do not use them.

They will not be removed during the regression-reproduction stage. Their presence is retained for provenance until the inherited baseline has been fully reproduced and subsequent cleanup can be separated from physics changes.

16.7. Full Regression Case Prepared

After the successful smoke test, the complete inherited regression case was generated through the same framework:

```
runs/formation_baseline_legacy_2000/
```

The generated case passed the framework submission integrity check.
The relevant active settings were confirmed as

```
my_constants.Nt = 2000  
my_constants.Np = 200  
diagnostics.diagnostics_names = diag1 chk  
chk.intervals = 2000
```

Thus, Framework v0.1 is now capable of reconstructing the full inherited 2000-step case and preparing it for submission.

The full regression was intentionally **not submitted** today.

Unlike the 20-step workflow test, the inherited case requires a substantially longer SCNet allocation and produces large diagnostic/checkpoint output. Running it is therefore deferred until its computational cost and the intended SCNet resource/quota policy are confirmed.

16.8. Status at End of Session

At the end of 2026-09-23, Framework v0.1 has reached the following state:

- structured formation configuration operational;
- configuration validation operational;
- derived-quantity checks operational;
- WarpX input generation operational;
- legacy active-key reproduction verified;
- checkpoint control operational;
- SCNet Slurm generation operational;
- provenance manifest and file-integrity checks operational;
- submission wrapper operational;
- 20-step end-to-end smoke test successfully completed;
- full 2000-step inherited regression generated and dry-run verified;
- full regression intentionally left unsubmitted.

The immediate next technical milestone is no longer basic framework execution. The next stage is to prepare the quantitative regression-comparison workflow and, when appropriate, execute the inherited 2000-step reference case through Framework v0.1 for comparison with the frozen legacy runs.

17. Research Log — 2026-09-24

The objective of today's session was to execute the complete inherited 2000-step formation case through Framework v0.1 and determine whether the new framework reproduces the frozen legacy simulations.

This was the first full physics regression after the successful 20-step workflow smoke test.

17.1. Full 2000-Step Framework Regression

The generated regression case

formation_baseline_legacy_2000

had previously passed configuration validation, input generation, provenance hashing, and the submission dry-run.

The active settings relevant to the inherited baseline were confirmed as

```
my_constants.Nt = 2000
my_constants.Np = 200
diagnostics.diags_names = diag1 chk
chk.intervals = 2000
```

The case was submitted to SCNet as Slurm job

```
122899629
```

and executed on node

```
i08r4n07
```

using

4 MPI ranks $\times$ 8 OpenMP threads/rank = 32 allocated CPU cores.

The job completed successfully:

```
State      = COMPLETED
Elapsed    = 02:36:33
ExitCode   = 0:0
```

The framework runtime record also reported

```
EXIT_CODE=0
```

and the total generated output occupied approximately

12 GB.

The output structure reproduced the inherited case:

- 21 plotfile diagnostics from step 0 to step 2000 in increments of 100;
- checkpoint at step 0;
- checkpoint at step 2000;
- `warpx_used_inputs`;
- runtime provenance information.

The wall-clock time was also essentially identical to the inherited runs, which required approximately 2 h 36 min.

This provided an initial indication that Framework v0.1 was executing the same numerical workload.

17.2. Scalar Regression Against the Legacy Runs

The new framework realization was compared against the two frozen inherited runs

```
legacy_i : 111646811
legacy_j : 111646815
framework: 122899629
```

using the same diagnostic definitions developed during the Week 2 baseline audit. The principal observables were

$$B_{z,\text{center}}, \quad B_{z,\text{min}}, \quad B_{z,\text{max}},$$

$$\rho_{\text{center}}, \quad \rho_{\text{max}},$$

and

$$T_{i,\text{center}}, \quad T_{i,\text{max}}.$$

Because the particle initialization uses `NRandomPerCell` and no explicit deterministic random seed had been identified, exact byte-for-byte agreement was not expected. The appropriate comparison is therefore the agreement of macroscopic observables between independent stochastic PIC realizations.

The maximum deviations of the framework realization from the arithmetic midpoint of the two legacy runs were approximately

$$0.082\% \quad \text{for} \quad B_{z,\text{center}},$$

$$0.71\% \quad \text{for} \quad B_{z,\text{min}},$$

$$0.23\% \quad \text{for} \quad B_{z,\text{max}},$$

$$0.38\% \quad \text{for} \quad \rho_{\text{center}},$$

and

$$1.73\% \quad \text{for} \quad \rho_{\text{max}}.$$

The ion-temperature diagnostics showed larger realization-to-realization variation:

$$2.89\% \quad \text{for} \quad T_{i,\text{center}},$$

while the pointwise chamber maximum reached a deviation of approximately

$$24.2\%.$$

The latter quantity is considered an informative diagnostic rather than a strict regression observable because pointwise kinetic-temperature maxima are particularly sensitive to PIC sampling noise.

Figure ?? shows representative temporal comparisons.

The interval bounded by the two legacy realizations was not interpreted as a statistical tolerance band. With only two inherited stochastic samples, whether a third realization lies numerically between them at a particular time is not a meaningful pass/fail criterion.

Instead, the magnitude and structure of the differences were examined.

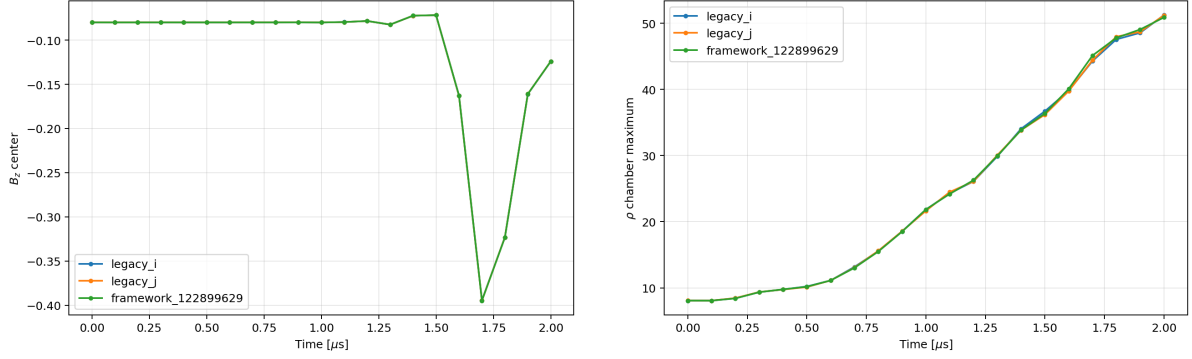


Figure 1: Scalar regression of Framework v0.1 against the two inherited stochastic realizations. Left: central axial magnetic field. Right: maximum proton charge-density diagnostic within the physical chamber. The framework realization follows the same macroscopic evolution as both legacy runs.

17.3. Spatial Regression

A second regression test compared the full midplane spatial structure at

$$t = 0, 1.0, 1.7, 2.0 \mu\text{s},$$

corresponding to simulation steps

$$0, 1000, 1700, 2000.$$

For the magnetic field, the relative spatial L_2 difference between the framework result and the arithmetic midpoint of the two legacy runs was

$$0 \quad \text{at} \quad t = 0,$$

$$2.83 \times 10^{-3} \quad \text{at} \quad t = 1.0 \mu\text{s},$$

$$2.71 \times 10^{-3} \quad \text{at} \quad t = 1.7 \mu\text{s},$$

and

$$5.19 \times 10^{-3} \quad \text{at} \quad t = 2.0 \mu\text{s}.$$

Thus, the spatial magnetic-field discrepancy remained below approximately

$$0.52\%$$

over the complete inherited simulation.

The corresponding spatial correlations were

$$0.999996, \quad 0.999996, \quad 0.999986$$

at 1.0, 1.7, and 2.0 μs , respectively.

The sign of B_z agreed over essentially the complete physical midplane:

$$100.000\% \quad \text{at} \quad 1.0 \mu\text{s},$$

$$99.991\% \quad \text{at} \quad 1.7 \mu\text{s},$$

and

$$99.982\% \quad \text{at} \quad 2.0 \mu\text{s}.$$

The simple $B_z = 0$ field-reversal indicator also evolved identically in all three simulations:

Time	legacy i	legacy j	framework
0.0 μs	0	0	0
1.0 μs	1	1	1
1.7 μs	1	1	1
2.0 μs	1	1	1

This indicator is used only as a field-reversal proxy and must not be interpreted as a formally reconstructed magnetic separatrix.

At the strongest compression stage, near

$$t \simeq 1.7 \mu\text{s},$$

the fraction of the analyzed midplane with negative B_z was

$$0.310973, \quad 0.311153, \quad 0.311153$$

for legacy i , legacy j , and Framework v0.1, respectively.

Figure ?? shows a representative magnetic profile comparison and the residual field at the end of the regression.

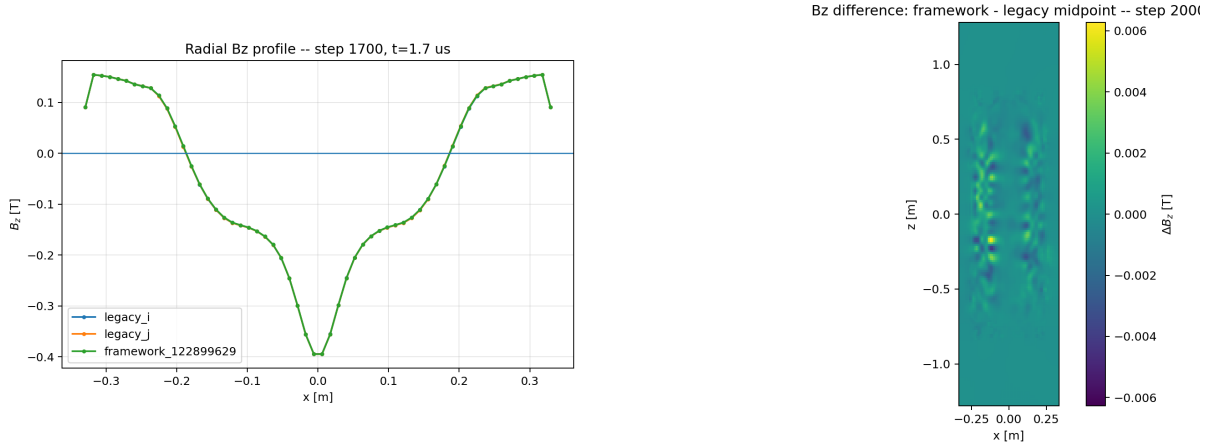


Figure 2: Spatial magnetic-field regression. Left: radial B_z profiles from both legacy realizations and Framework v0.1 near the strongest compression stage at $t = 1.7 \mu\text{s}$. Right: residual B_z field between the framework realization and the arithmetic midpoint of the two legacy runs at $t = 2.0 \mu\text{s}$.

The proton-density field showed similarly strong structural agreement. Its relative spatial L_2 differences were approximately

$$0.24\%, 0.79\%, 1.82\%, 3.68\%$$

at 0, 1.0, 1.7, and 2.0 μs , respectively.

Even at 2.0 μs , the spatial density correlation remained

$$0.99907.$$

The ion-temperature field accumulated substantially larger stochastic differences, reaching a relative spatial L_2 difference of approximately

$$25.6\%$$

at $2.0 \mu s$, while retaining a spatial correlation of approximately

$$0.953.$$

This behavior is consistent with the larger sensitivity of kinetic temperature diagnostics to finite macroparticle sampling and reinforces the decision not to use pointwise T_i extrema as strict regression criteria.

17.4. Framework v0.1 Regression Result

The complete evidence now includes

- successful structured configuration generation;
- reproduction of all 111 active WarpX input assignments;
- identical mesh, timestep, solver configuration, and field drive;
- successful SCNet MPI/OpenMP execution;
- essentially identical wall-clock runtime;
- identical diagnostic and checkpoint structure;
- sub-percent magnetic-field differences in the scalar diagnostics;
- less than approximately 0.52% relative spatial L_2 difference in B_z ;
- essentially identical field-reversal geometry;
- strong spatial agreement of the proton-density field;
- remaining differences concentrated primarily in stochastic kinetic-temperature diagnostics.

On this basis, the inherited-baseline reproduction is recorded as

Framework v0.1 inherited-baseline regression: PASS

Here, “PASS” denotes successful reproduction of the inherited macroscopic model and numerical behavior. It does not imply that a statistically rigorous stochastic tolerance interval has been established, since only two frozen legacy particle realizations are currently available.

17.5. End of the Takeover/Reproduction Stage

The original formation case has now been

audited $\rightarrow$ parameterized $\rightarrow$ generated $\rightarrow$ executed $\rightarrow$ regression tested.

Therefore, additional repetitions of the same inherited 2000-step formation case are not presently justified.

The validated baseline can now serve as a frozen regression reference while future framework development proceeds without modifying the verified legacy configuration.

The next project stage will move from infrastructure reproduction toward the scientific problem prioritized for the short term: reproduction and audit of the Grad-Shafranov/MRR initialized FRC and its three-dimensional tilt instability.

The main unresolved quantities to investigate in that stage include

- the inherited Alfvén-time normalization;
- the definition and implementation of S^* ;
- consistency of proton versus deuterium mass in the ion skin depth;
- the definition of the tilt-mode growth rate γ_{tilt} ;
- interpretation of the inherited velocity/current diagnostic;
- reproduction of seeded and unseeded $n = 1$ tilt cases.

This marks the end of the initial software-takeover and inherited-formation reproduction phase.